INTERFACE

In computing, an interface is a shared boundary across which two or more separate components of a computer system exchange information. The exchange can be between software, computer, hardware, humans, and combinations of these. Some computer hardware devices, such as a touch-screen, can both send and receive data through the interface, while others such as keyboard or microphone may only provide an interface to send data to a given system.

HARDWARE INTERFACE

Hardware interfaces exist in many components, such as the various buses, storage devices, other input/output (I/O) devices, etc. a hardware interface is described by the mechanical, electrical, and logical signals at the interface, and the protocol for sequencing them. Hardware interfaces can be **parallel** with several electrical connections carrying parts of the data simultaneously or **serial** where data are sent one bit at a time.

SOFTWARE INTERFACE

A software interface may refer to a wide range of difference types of interface at different “levels”: an operating system may interface with pieces of hardware. Applications or programs running on the operating system may need to interact via data streams, filters, and pipelines; and in object-oriented programs, objects within an application may need to interact via method.

USER INTERFACES

A user interface is a point of interaction between a computer and humans; it includes any number of modalities of interaction (such as graphics, sound, position, movement, etc.) where data is transferred between the user and the computer System.

I/O INTERFACE

I/O interface is the process of transferring of information between internal and external storage.

Internal storage includes CPU and memory. External storage includes all the peripheral devices which are connected outside the computer.

The communication between the CPU, memory and peripheral devices is handled by interfacing circuit called I/O interface. In order to interface between the CPU and memory a special communication lines are required, which resolves the differences between the peripherals.

I/O BUSES AND INTERFACE MODULE

These synchronizes the data flow and supervised the transfer of information between the CPU and peripherals. A simple I/O buses and the interface module is as shown below;

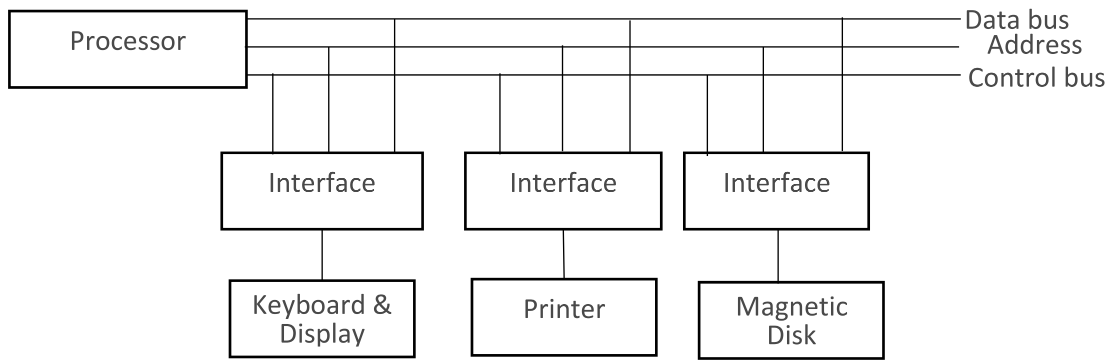


Figure 4.0: I/O Buses and interface module

Data Bus: - Hold the data

Address bus: - hold the address of the data

Control bus: - pass the control signal

TYPES OF COMMANDS IN I/O BUS AND INTERFACE MODULE

1. Control command: issued command to activate the peripheral and to inform it what to do.
2. Status: test various status conditions in the peripheral circuit to know whether it is ready or not.
3. Data output command: transferring data from bus into one of its register
4. Data input command: transferring data from register to one of the buses.

Example of I/O Interface Circuit

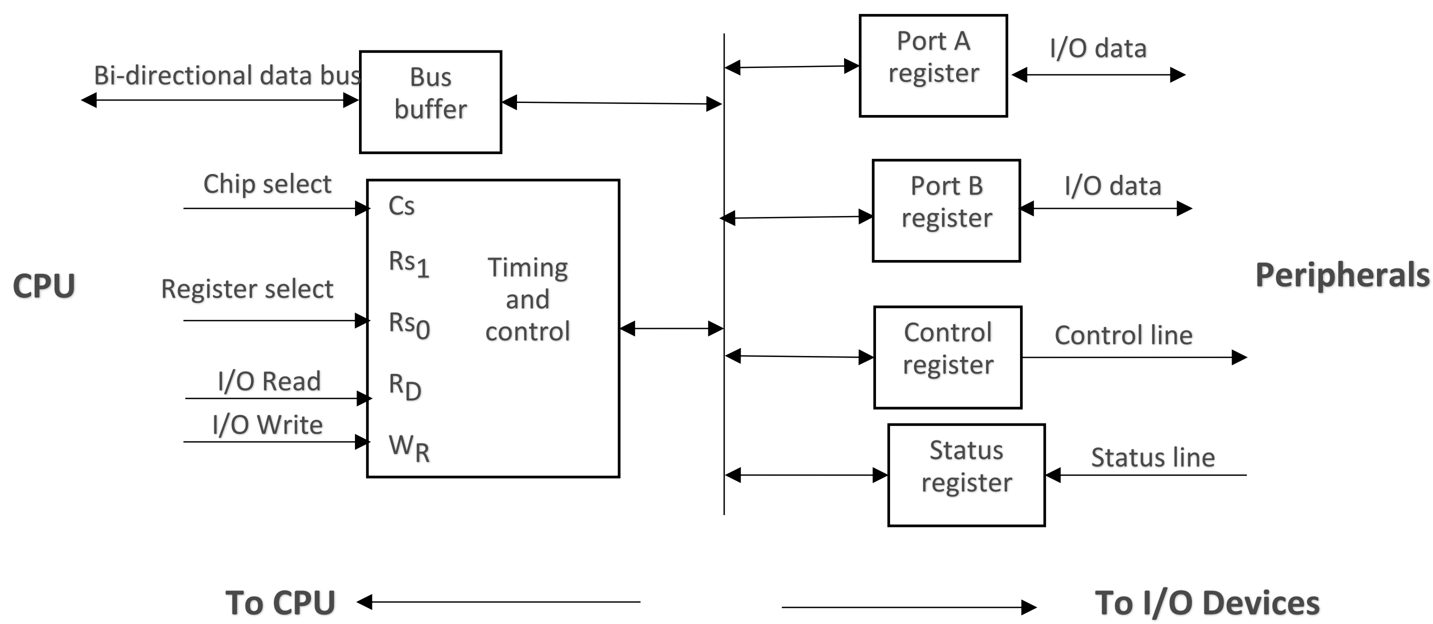


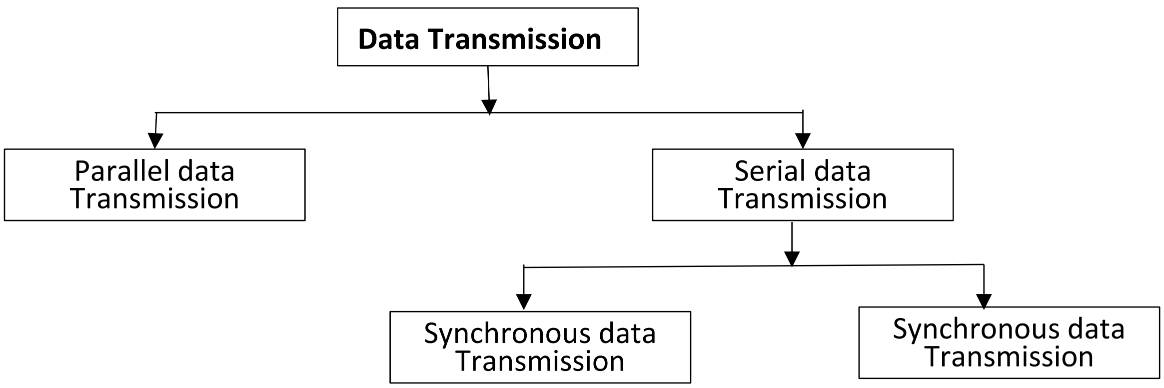
Figure 4.1: I/O Interface circuit

Cs	Rs_1	Rs_0	
0	X	X	None: Data bus in high-impedance state
1	0	0	Port A register
1	0	1	Port B register
1	1	0	Control register
1	1	1	Status register

DATA TRANSFER

Data transfer instruction transfer data between memory and processor registers, processor register and I/O devices and from one processor register to another.

There are two types of data transfer: Parallel data transfer and serial data transfer. Serial data transfer is further divided into two; synchronous and asynchronous data transfer. These can be illustrated in the chart below.



Parallel Data Transfer

Parallel data transmission is a transmission structure that shows multiple data bits at a similar time over a separate media.

In general, parallel transmission can be used with wired channel that uses multiple separate wires. For example, if you want to transmit 1 byte of data using parallel transmission, you need 8 wires to do that as data are transmitted in bits.

Here, 8 bits are sent together at a time which require 8 lines or wires. Example of parallel data transfer is the data transfer between computer and printer.

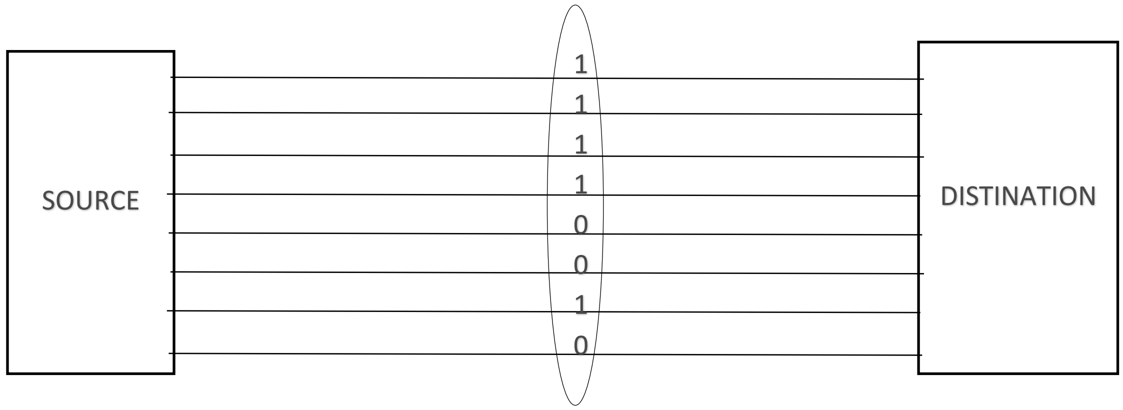


Figure 4.2: Parallel data transmission

Advantages of Parallel data transfer

I. Data transfer rate is very high

II. Huge amount of data is transferred over connection lines.

Disadvantages of Parallel data transfer

I. It is usually limited to shorter distance

II. As the number of bits are increasing the cost and the complexity of the system also increases

Serial Data Transfer

Here all bits of the byte are transmitted one after the other on the same wire. With the importance of speed, it may seem that anyone designing a data communication system would choose parallel transmission. However, most communication system uses serial mode. Example of serial transmission is the transmission between Computer and computer.

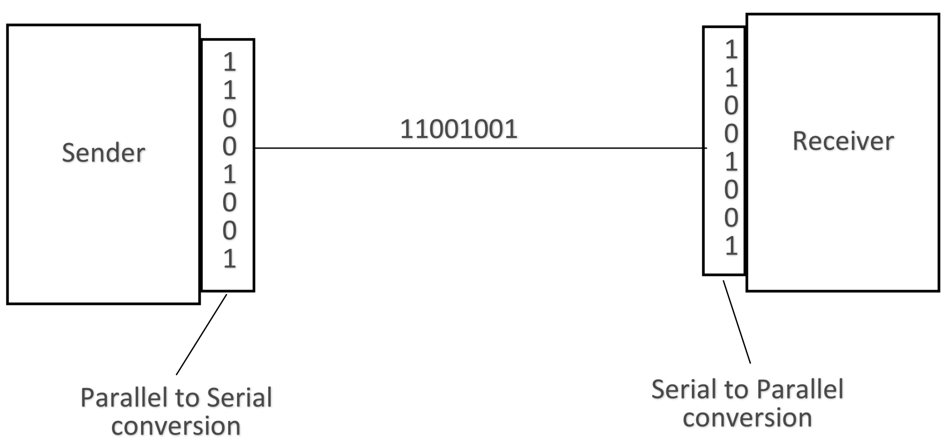


Figure 4.3: Serial Data Transmission

Types of Serial Data transfer

1) Synchronous data transfer:
   Synchronous data transmission means at a time, the bytes are transmitted as a block in a continuous stream of bits
2) No start and stop bits are used
3) It does not care whether the data is received by the receiver or not.

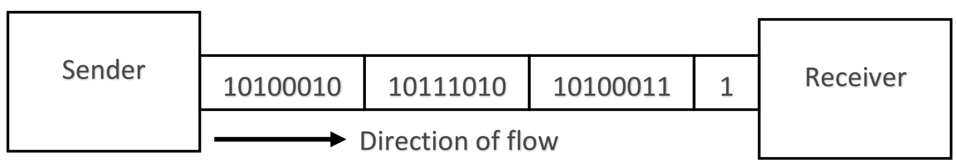


Figure 4.4: Synchronous data Transmission

Asynchronous Data transfer

Asynchronous Data transfer means "at a regular interval". Here, the transmitter transmits data bytes at any instance of time and only 1 byte is sent at a time. There is ideal time between two data bytes.

Transmitter and receiver operate at different clock frequencies, to help the receiver start and stop bits are used along with data in middle. By seeing the stop bit it assume that 1 byte of data is received. Timing of signal is not important.

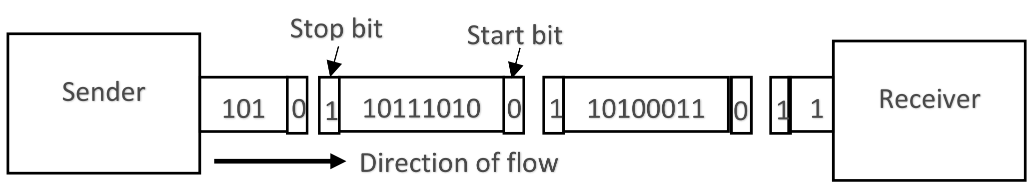


Figure 4.5: Asynchronous data Transmission

Advantages of Serial data transfer

1) Serial networks can be continued over long distances at much less cost because fewer physical wires are required, and intermediate electronics elements are less cheap.
2) Using only one physical wire defines that there are never timing issues caused by one wire being slightly higher than the other.

Disadvantages of Serial data transfer

1) Data transfer rate is very slow compare to parallel transmission.
2) Use of conversion devices at the source and destination end may lead to increase in overall transmission.

COMPUTER INTERRUPT

An Interrupt in computer architecture is a signal that requests the processor to suspend its current execution and service the occurred interrupt.

To service the interrupt, the processor executes the corresponding interrupt service routine (ISR). After the execution of the ISR, the processor resumes the execution of the suspended program.

Interrupt techniques used in I/O interface

1) Programmed I/O interface
2) Interrupt driven I/O
3) Direct Memory Access (DMA)

These techniques are used to transfer data between user and the CPU.

Programmed I/O

This technique is the simplest to exchange data between external devices and the processors.in this technique, the processor or the central processing unit runs or executes a program giving direct control of I/O operations.

Processor issues a command to I/O module and waits for the operation to complete. Also, the processor keeps checking the I/O module status until if find the completion of the operation. The processor’s time is wasted, in case the processor is faster than the I/O module. Its module is considered to be a slow module. The cycle of the operation is as shown in the flow chart below.

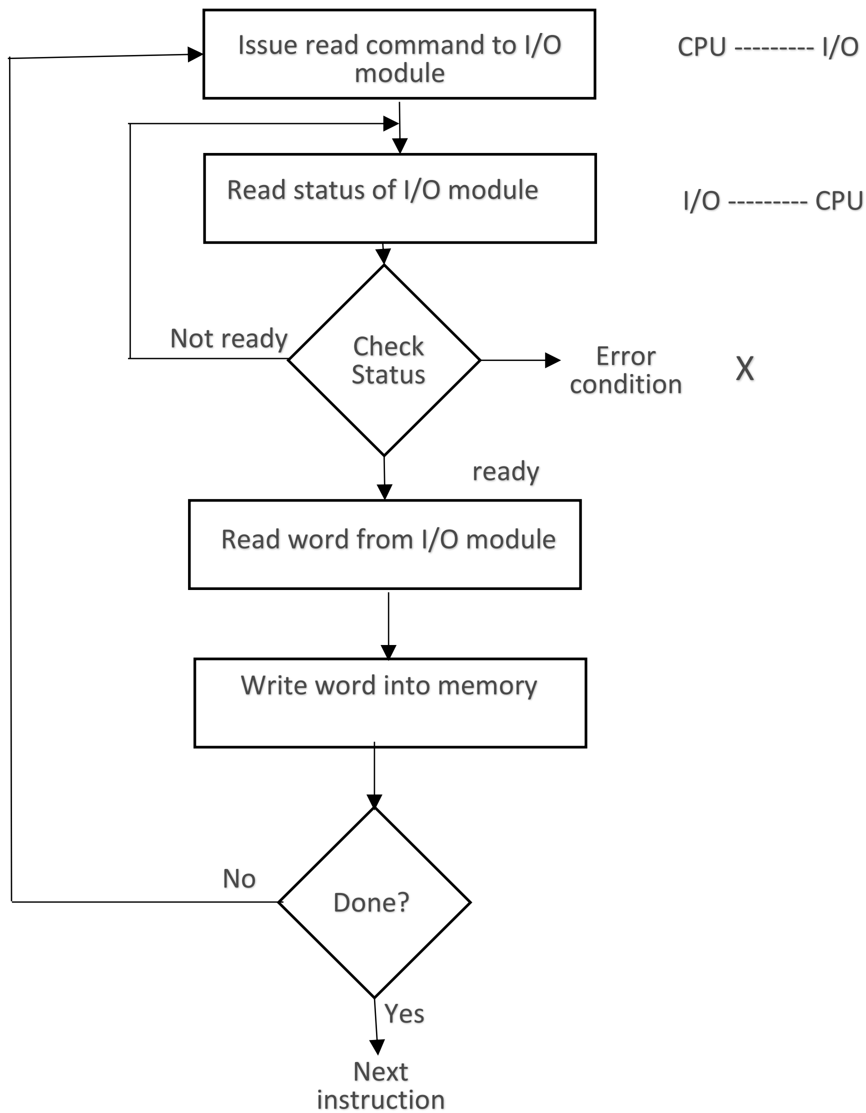


Figure 4.6: Flow chart illustrating Programmed I/O interrupt

INTERRUPT DRIVEN I/O

It is similar to Programmed I/O.

Here, the processor issues an I/O command to a module and then go on to do some other useful work. The I/O module will then interrupt the processor to request service, when it is ready to exchange data with the processor. The processor then executes the data transfer as before and then resumes its former processing. Interrupt-driven I/O still consumes a lot of time because every data has to pass with processor.

Basic operation of Interrupt-driven I/O

i. CPU issues read command
ii. I/O module gets data from Peripheral whilst CPU does other work
iii. I/O module interrupt CPU
iv. CPU request data

v. I/O module transfers data

Interrupt driven I/O processing Cycle

i. A device driver initiates an I/O request on behalf of a processor
ii. The device driver signals the I/O controller for the proper device, which initiates the requested I/O
iii. The device signals the I/O controller that is ready to receive input, the output is complete or that an error has been generated.
iv. The CPU receives the interrupt signal on the interrupt-request line and transfer control to the interrupt handler routine.
v. The interrupt handler determines the cause of the interrupt, performs the necessary processing and executes a return from interrupt instruction.
vi. The CPU returns to the execution state prior to the interrupt being signaled.
vii. The CPU continues processing until the cycle begins again.

The cycle is as shown below:

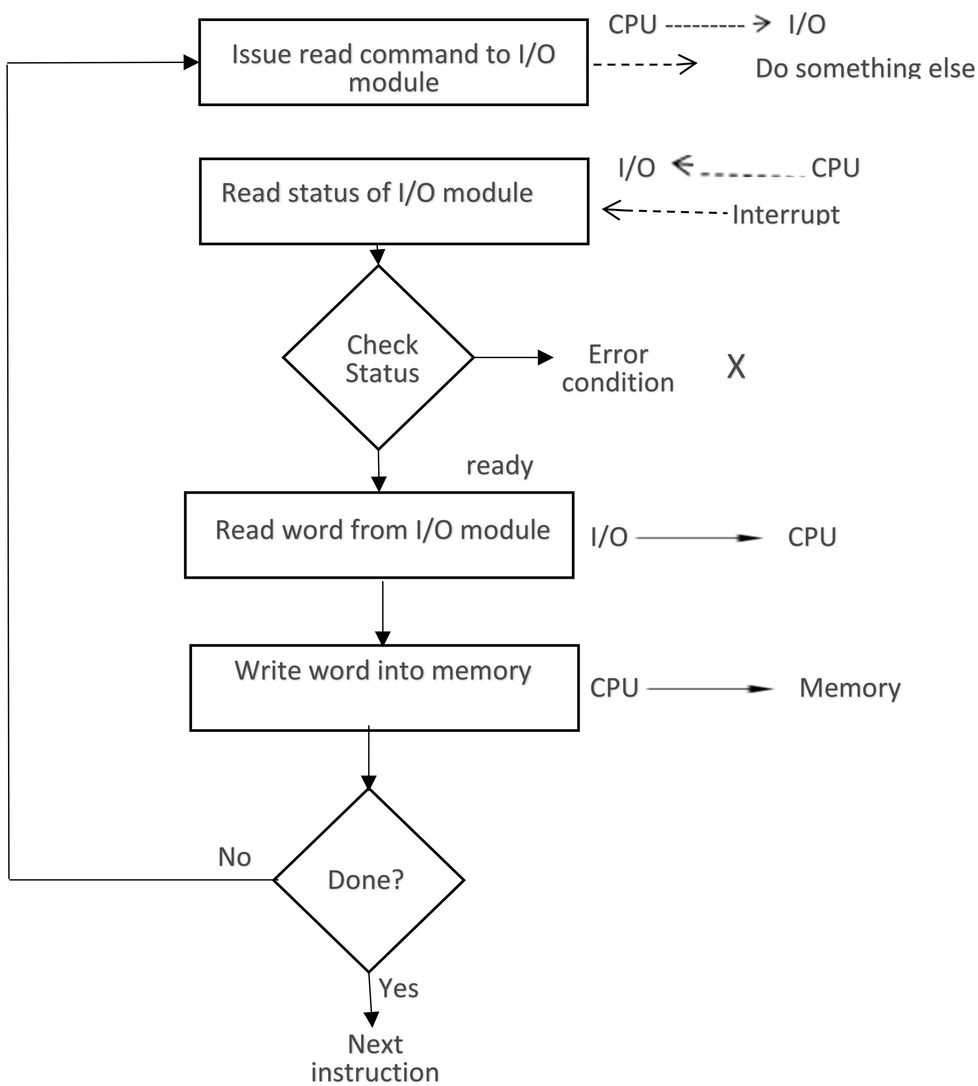


Figure 4.7: Flow chart illustrating Interrupt driven I/O

Direct Memory Access (DMA)

Direct memory access (DMA) is a mode of data transfer between the memory and I/O devices. This happen without the involvement of the processor.

DMA Controller

DMA controller is a hardware unit that allows I/O devices to access memory directly without the participation of the processor.

Below is the diagram of a DMA controller

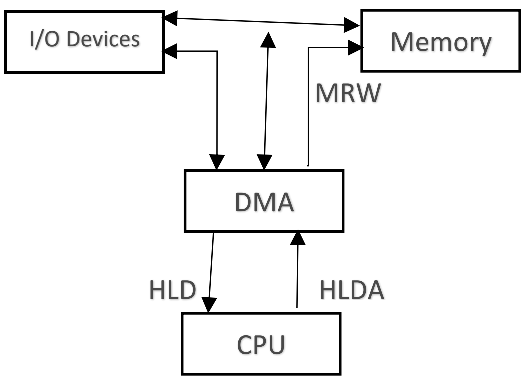


Figure 4.8: DMA Controller I/O

DMA controller Working

I. Whenever an I/O device wants to transfer the data to or from memory, it sends the DMA request (DRQ) to DMA controller. DMA controller accept the DRQ and ask CPU to hold a few clock cycles by sending it the Hold request (HLD).

II. CPU receives the HLD from DMA controller and relinquishes the bus and sends the Hold acknowledgement (HLDA) to DMA controller.

III. After receiving the HLDA, DMA controller acknowledges I/O device (DACK) that the transfer can be performed and DMA controller takes the charge of the system bus and transfers the data to or from memory.

IV. When the data transfer is accomplished the DMA raise an interrupt to let know the processor that the task of data transfer is finished and the processor can take control over the bus again and start processing where it has left.

Single Chip and Multi-Chip Microprocessor

Single-Chip Microprocessor:

Putting a CUP, ROM, RAM and data I/OA circuitry into a single IC will make a single-chip microprocessor. Single-chip come in compact and cheap, but do not allow users to choose built-in functions at their option. Single-chip microprocessors are also called microcomputer units (MCUs) because they are made up of single IC.

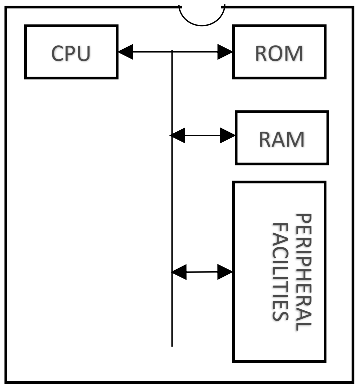


Figure 4.9: Single-chip microprocessor

Multi-chip microprocessor

On the other hand, a computer fabricated from a mix of a CPU, Memory and data I/O devices is called a multi-chip microprocessor. Multi-chip microprocessors offer users greater freedom in their component choice. Multi-chip microprocessor will prove more advantageous for larger systems involving complexities.

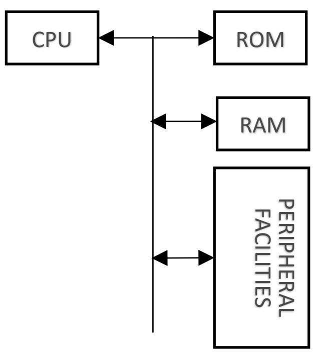


Figure 4.10: Multi-chip microprocessor

EVOLUTION OF IC FROM SSI, MSI, LSI, to VLSI

1. Small Scale Integration (SSI): firstly, between 1961 to 1965, in SSI transistors were fabricated about 10 to 100 on a single chip. This technology is used for making logic gates, flip-flops.
2. Medium Scale Integration (MSI): in 1966 to 1970 in MSI transistors were fabricated about 100 to 1000 on a single-chip. This technology is used for making counters, multiplexers, decoders.
3. Large Scale Integration (LSI): in 1971 to 1979 in LSI transistors were fabricated about 1000 to 20,000 on a single-chip. This technology is used for making microprocessor, RAM, ROM.
4. Very Large-Scale Integration (VLSI): in 1980 to 1984 in VLSI transistors were fabricated about 20000 to 50000 on a single-chip. This technology is used for making Digital Signal Processing (DSP), ICs, RISC microprocessor, 16-bit and 32-bit microprocessors.
5. Ultra-Large-Scale Integration (ULSI); from 1985 to present technology ULSI, in which transistors are fabricated about greater than 50000 to billions on a single chip. This technology is used for making 64 bit microprocessor.